

14.4

DATA
SET

DS1406

10. Marriage Rates Half of adults today are married compared with the all-time high of 72% in 1960.⁹ Do marriage rates vary by educational level? A sample of 300 adults in each of three educational categories provided the following information.

Educational Level	Married	
	Yes	No
Bachelors or higher	187	113
Some college	162	138
High school or less	149	151

 $n = 900$

498 | 402

Test to determine whether marriage rates differ significantly among adults in these educational levels using $\alpha = .01$. How would you summarize your results?

213母体是否差异

E_{ij}	Yes	No
1	$\frac{187}{900} \times 300$	$\frac{113}{900} \times 300$
2	166	134
3	166	134

$$\textcircled{1} \begin{cases} H_0: p_1 = p_2 = p_3 \\ H_a: p_i \neq p_j \ (i \neq j) \end{cases}$$

$$\textcircled{2} \alpha = 0.01$$

$$\textcircled{3} \chi^2_{STAT} = \sum_{i=1}^3 \sum_{j=1}^2 \frac{(O_{ij} - E_{ij})^2}{E_{ij}} \sim \chi^2_{(3-1)(2-1)} = 2$$

$$\textcircled{4} \chi^2 = \frac{(187-166)^2}{166} + \frac{(113-134)^2}{134} + \frac{(162-166)^2}{166} + \frac{(138-134)^2}{134} + \frac{(149-166)^2}{166} + \frac{(151-134)^2}{134} \approx 10.06$$

$$\textcircled{5} RR: \chi^2_{2, 0.01} = 9.21$$

$\textcircled{6}$ reject H_0 : differ significantly.

14.4

DATA
SET

DS1407

12. How Big Is the Household? A local chamber of commerce surveyed 120 households in their city—40 in each of three types of residence (apartment, duplex, or single residence)—and recorded the number of family members in each of the households. The data are shown in the table.

Family Members	Type of Residence			Ward
	Apartment	Duplex	Single Residence	
1	8	20	1	29
2	16	8	9	33
3	10	10	14	34
4 or More	6	2	16	24
	40	40	40	

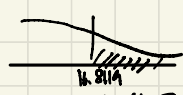
Is there a significant difference in the family size distributions for the three types of residence? Test using $\alpha = .01$. If there are significant differences, describe the nature of these differences.

$$\textcircled{1} \begin{cases} H_0: p_1 = p_2 = p_3 \\ H_a: p_i \neq p_j \ (i \neq j) \end{cases}$$

$$\textcircled{2} \alpha = 0.01$$

$$\textcircled{3} \chi^2_{STAT} = \sum_{i=1}^r \sum_{j=1}^c \frac{(O_{ij} - E_{ij})^2}{E_{ij}} \sim \chi^2_{(r-1)(c-1)} = 6$$

$$\textcircled{4} \chi^2_* = \frac{(8-9.67)^2}{9.67} + \frac{(16-11)^2}{11} + \dots + \frac{(16-8)^2}{8} = 36.4929$$

$$\textcircled{5} \chi_{0.01, 6} = 16.8119$$


$$\textcircled{6} \therefore \text{reject } H_0: \text{分佈有顯著差異}$$

n=120

 E_{ij}

$\frac{29}{120} \times 40 = 9.67$	9.67	9.67
$\frac{33}{120} \times 40 = 11$	11	11
$\frac{34}{120} \times 40 = 11.33$	11.33	11.33
$\frac{24}{120} \times 40 = 8$	8	8

13.1

12. Dissolved Oxygen Content Dissolved oxygen content is a measure of the ability of a river, lake, or stream to support aquatic life, with high levels being better. A pollution-control inspector who suspected that a river community was releasing semitreated sewage into a river, randomly selected five specimens of river water at a location above the town and another five below. These are the dissolved oxygen readings (in parts per million):

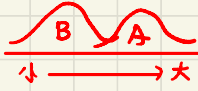
Above Town	4.8	5.2	5.0	4.9	5.1
Below Town	5.0	4.7	4.9	4.8	4.9

- Use a one-tailed Wilcoxon rank sum test with $\alpha = .05$ to confirm or refute the theory.
- Use a Student's t -test (with $\alpha = .05$) to analyze the data. Compare the conclusion reached in part a.

Type	B	A	B	A	B	B	A	B	A	A
Data	4.7	4.8	4.8	4.9	4.9	4.9	5.0	5.0	5.1	5.2
Rank	1	2.5	2.5	5	5	5	7.5	7.5	9	10

$\rightarrow \text{Total} = 55 = \frac{(1+10) \cdot 10}{2}$ (驗算)

① $\begin{cases} H_0 = & A, B \text{ 分佈一樣} \\ H_a = & B \text{ 分佈於 } A \text{ 分佈左側} \end{cases}$ 左尾

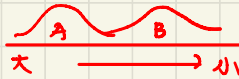


$\therefore$ 看夠不夠小.

$$T_1 = 34$$

$$T_1^* = 5 \times (5+5+1) - 34 = 21$$

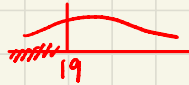
看 T_1^* 夠不夠小



② $\alpha = 0.05$

③ $\text{STAT} = T_1^* - 21$

④ $T_{0.05} = 19$



⑤ T_1^* 不夠小, 無法反轉 $A > B$

$\therefore$ non-reject H_0 .

15.1

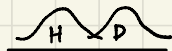
DATA
SET

DS1502

13. Eye Movement In an investigation of the visual scanning behavior of deaf children, measurements of eye movement were taken on nine deaf and nine hearing children. The table gives the eye-movement rates and their ranks (in parentheses). Does it appear that the distributions of eye-movement rates for deaf children and hearing children differ?

Deaf Children	Hearing Children
2.75 (15)	.89 (1)
2.14 (11)	1.43 (7)
3.23 (18)	1.06 (4)
2.07 (10)	1.01 (3)
2.49 (14)	.94 (2)
2.18 (12)	1.79 (8)
3.16 (17)	1.12 (5.5)
2.93 (16)	2.01 (9)
2.20 (13)	1.12 (5.5)
Rank Sum	126
	45

$$\textcircled{1} \begin{cases} H_0 = H = D \\ H_a = H \neq D \end{cases}$$



$$\textcircled{2} \begin{aligned} T_1 &= 45 \\ T_1^* &= (9 + 9 + 1) \times 9 - 45 = 126 \\ \min(45, 126) &= 45 \end{aligned}$$

$$\textcircled{3} \alpha = 0.05$$

$$\textcircled{4} T_{0.05} = 62$$



$\textcircled{5}$ reject H_0 : differs !

15.2

Two Simple Examples Use the sign test to compare two populations for significant differences for the paired data in Exercises 4–5. State the null and alternative hypotheses to be tested. Determine an appropriate rejection region with $\alpha \leq .10$. Calculate the observed value of the test statistic and present your conclusion.

4.

Population	Pair						
	1	2	3	4	5	6	7
1	8.9	8.1	9.3	7.7	10.4	8.3	7.4
2	8.8	7.4	9.0	7.8	9.9	8.1	6.9
	⊕	⊖	⊕	⊖	⊕	⊕	⊕

双尾

① $H_0: p = 0.5$
 $H_a: p \neq 0.5$ ② $\alpha = 0.01$ ③ test statistic $X = \#(+), n = 7$

④ $X = 6$ ⑤ $p\text{-value} = 2 \cdot P(X \geq 6) = 0.125 > \alpha = 0.1$

$$P(X \geq 6) = C_6^7 \cdot (0.5)^6 (0.5)^1 + C_7^7 \cdot (0.5)^7 = 0.0625$$

= 二项分布 $n=7$ $p=0.5$

⑥ $\therefore$ non-reject H_0 : no significant difference.

DATA SET
DS1507

11. Gourmet Cooking Two chefs, A and B, rated 22 meals on a scale of 1–10. The data are shown in the table. Do the data provide sufficient evidence to indicate that **one of the chefs tends to give higher ratings than the other?** Test by using the sign test with a value of α near .05.

Meal	A	B	Meal	A	B
1	6	8	12	8	5
2	4	5	13	4	2
3	7	4	14	3	3
4	8	7	15	6	8
5	2	3	16	9	10
6	7	4	17	9	8
7	9	9	18	4	6
8	7	8	19	4	3
9	2	5	20	5	4
10	4	3	21	3	2
11	6	9	22	5	3

- a. Use the binomial tables in Appendix I to find the exact rejection region for the test.
- b. Use the **large-sample z statistic**. (NOTE: Although the large-sample approximation is suggested for $n \geq 25$, it works fairly well for values of n as small as 15.)
- c. Compare the results of parts a and b.

■ **Table 1 Cumulative Binomial Probabilities**
Tabulated values are $P(x \leq k) = p(0) + p(1) + \dots + p(k)$.
(Computations are rounded to the third decimal place.)

n = 20		P																		
k	.01	.05	.10	.20	.30	.40	.50	.60	.70	.80	.90	.95	.99	k						
0	.818	.358	.122	.012	.001	.000	.000	.000	.000	.000	.000	.000	.000	0						
1	.983	.736	.392	.069	.008	.001	.000	.000	.000	.000	.000	.000	.000	1						
2	.999	.925	.677	.206	.035	.004	.000	.000	.000	.000	.000	.000	.000	2						
3	1.000	.984	.867	.411	.107	.016	.001	.000	.000	.000	.000	.000	.000	3						
4	1.000	.997	.957	.630	.238	.051	.006	.000	.000	.000	.000	.000	.000	4						
5	1.000	1.000	.989	.804	.416	.126	.021	.002	.000	.000	.000	.000	.000	5						
6	1.000	1.000	.998	.913	.608	.250	.058	.006	.000	.000	.000	.000	.000	6						
7	1.000	1.000	1.000	.968	.772	.416	.132	.021	.001	.000	.000	.000	.000	7						
8	1.000	1.000	1.000	.990	.887	.596	.252	.057	.005	.000	.000	.000	.000	8						
9	1.000	1.000	1.000	.997	.952	.755	.412	.128	.017	.001	.000	.000	.000	9						
10	1.000	1.000	1.000	.999	.983	.872	.588	.245	.048	.003	.000	.000	.000	10						
11	1.000	1.000	1.000	1.000	.995	.943	.748	.404	.113	.010	.000	.000	.000	11						
12	1.000	1.000	1.000	1.000	.999	.979	.868	.584	.228	.032	.000	.000	.000	12						
13	1.000	1.000	1.000	1.000	1.000	.994	.942	.750	.392	.087	.002	.000	.000	13						
14	1.000	1.000	1.000	1.000	1.000	.998	.979	.874	.584	.196	.011	.000	.000	14						
15	1.000	1.000	1.000	1.000	1.000	1.000	.994	.949	.762	.370	.043	.003	.000	15						
16	1.000	1.000	1.000	1.000	1.000	1.000	.999	.984	.893	.589	.133	.016	.000	16						
17	1.000	1.000	1.000	1.000	1.000	1.000	1.000	.996	.965	.794	.323	.075	.001	17						
18	1.000	1.000	1.000	1.000	1.000	1.000	1.000	.999	.992	.931	.608	.264	.017	18						
19	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	.999	.988	.878	.642	.182	19						
20	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	20						

a. $\begin{cases} H_0: p = 0.5 \\ H_a: p \neq 0.5 \end{cases}$ 2x15 $\alpha = 0.05$

2. $P(X \leq 5) = 0.42$ 符合
2. $P(X \leq 6) = 1.16$ 不符合.

20 - 5 = 15
 $\begin{cases} \oplus 1, 2, 3, 4, 5 \\ \ominus 1, 2, 3, 4, 5 \rightarrow \oplus 15, 16, 17, \dots, 20: 2 \end{cases}$

$\therefore RR: \{ X \leq 5 \text{ 或 } X \geq 15 \}$
 $\therefore \text{non-reject } H_0$

b. = 二項分佈 $\longrightarrow$ 常態分佈
($np, n \times p \times (1-p)$)

"
(10, 5) $\because \mu = 10$
 $\sigma = \sqrt{5} \approx 2.236$

$$\bar{z} = \frac{x - \mu}{\sigma} = \frac{11 - 10}{2.236} = 0.447.$$

RR: $\frac{x - 10}{2.236} < -z_{0.025} \rightarrow x =$
 $x < 5.6173$
 $\frac{x - 10}{2.236} > z_{0.025}$
 $x > 14.3827$

c. $\therefore$ 是整數 $\therefore$ 一樣
 $RR: \{ x = 0, 1, 2, 3, 4, 5, 15, 16, 17, 18, 19, 20 \}$

15.2

DATA
SET

DS1508

13. Recovery Rates Clinical data concerning the effectiveness of two drugs in treating a particular condition (as measured by recovery in 7 days or less) were collected from 10 hospitals. You want to know **whether the data present sufficient evidence to indicate a higher recovery rate for one of the two drugs.**

a. Test using the sign test. Choose your rejection region so that α is near .05.

b. Why might it be inappropriate to use the Student's t -test in analyzing the data?

Hospital	Drug A		
	Number in Group	Number Recovered (7 days or less)	Percentage Recovered
1	84	63	75.0
2	63	44	69.8
3	56	48	85.7
4	77	57	74.0
5	29	20	69.0
6	48	40	83.3
7	61	42	68.9
8	45	35	77.8
9	79	57	72.2
10	62	48	77.4

Hospital	Drug B		
	Number in Group	Number Recovered (7 days or less)	Percentage Recovered
1	96	82	85.4
2	83	69	83.1
3	91	73	80.2
4	47	35	74.5
5	60	42	70.0
6	27	22	81.5
7	69	52	75.4
8	72	57	79.2
9	89	76	85.4
10	46	37	80.4

a

 $n=10$

1	2	3	4	5	6	7	8	9	10
-	-	+	-	-	+	-	-	-	-

$$\textcircled{1} \begin{cases} H_0: p = 0.5 \\ H_a: p \neq 0.5 \end{cases}$$

$\textcircled{2}$ test statistic: $\#(+), n=10$

$$\textcircled{3} x=2 \quad \boxed{2 < 10 \times 0.5}$$

$$\textcircled{4} p\text{-value} = 2 \times P(x \leq 2) = 2 \times (0.5^{10} + C_{10}^1 0.5^{10} + C_{10}^2 0.5^{10}) \\ \approx 2 \times 0.055 = 0.11 > 0.05 \quad \text{non-reject } H_0$$

$$p\text{-value} = 2 \cdot P(x \leq 1) = 2 \cdot 0.011 = 0.022$$

$$2 \cdot P(x \geq 2) = 0.11 > 0.05$$

$$\therefore 1 \text{ \& } 10-1=9$$

$$RR: \{x \leq 1 \text{ or } x \geq 9\}$$

$\textcircled{5}, x=2$ 不在RR内

$\therefore$ non-reject H_0

b.

分布: 团体要常態

$n=10$
p
.50
. .001
. .011
. .055
. .172
. .377
. .623
. .828
. .945
. .989
. .999
. 1.000

ϕ (a)

Pair	1	2	3	4	5	6	7	8	9	10
sign	-	-	+	-	-	+	-	-	-	-

1. $H_0: p = 0.5 \quad H_a: p \neq 0.5$

2. $\alpha = 0.1$

3. test statistic: $x = \#(+), n = 10$

4. realized statistic: $x = 2$

5. $p\text{-value} = 2 \times P(x \leq 2) = 2 \times 0.055 = 0.11 > \alpha = 0.1$

6. Because $p\text{-value} = 0.124 > \alpha$, H_0 is not rejected, which means that there is no sufficient evidence to indicate a higher recovery rate for one of the two drugs.

ϕ (b)

If the data are not the normal distribution, the Student's t -test may give the misleading results. Therefore, it is inappropriate to use the Student's t -test in analyzing the data that are non-normal distribution.